Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. Consider the following matrix: $A = \begin{bmatrix} 11 & \frac{1}{2} & -2 & 3 \\ 4 & 1 & 1 & 0 \end{bmatrix}$.

   (a) What type of matrix is A? Is a 2 by 4 matrix (2×4), so it's a rectangular matrix

   (b) What are the entries of A at positions $(1, 2)$ and $(2, 1)$?

   - $a_{1,2} = \frac{1}{2}$
   - $a_{2,1} = 4$

   (c) At which positions in A is the entry equal to 1? In $a_{2,2}$ and $a_{2,3}$

2. Write the matrix $\left(A = [a_{ij}]_{3\times 3}\right)$ defined by:

   (a) $a_{ij} = i + j$

   $$A = \begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}$$

   (b) $a_{ij} = \begin{cases} 0, & \text{if } i = j, \\ -1, & \text{if } i > j, \\ 2, & \text{if } i < j. \end{cases}$

   $$A = \begin{pmatrix} 0 & 2 & 2 \\ -1 & 0 & 2 \\ -1 & -1 & 0 \end{pmatrix}$$

3. For each of the following parts, determine, if possible, the real numbers x and y such that the matrices A and B are equal.

   Remember: Two matrices are equal if they have the same dimensions and all their corresponding entries **are equal**.

   (a) $A = \begin{bmatrix} 1 & 0 \\ x & y \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ \frac{y}{2} & x+1 \\ -1 & 3 \end{bmatrix}$

   Solution:

- From the first row: $a_{11} = 1 = b_{11}$ and $a_{12} = 0 = b_{12}$.
- From the second row: $a_{21} = x = \frac{y}{2} = b_{21}$ and $a_{22} = y = x + 1 = b_{22}$.
- From the third row: $a_{31} = -1 = b_{31}$ and $a_{32} = 3 = b_{32}$.

The point is that rows 1 and 3 do not give us any information about x and y, so we only need to solve the system of equations from the second row:

$$\begin{cases} x = \dfrac{y}{2}, \\ y = x + 1. \end{cases}$$

Substitute $x = \frac{y}{2}$ into the second equation:

$$\begin{aligned} y &= \frac{y}{2} + 1, \\ 2y &= y + 2, \\ y &= 2, \qquad x = \frac{y}{2} = 1. \end{aligned}$$

$$x = 1, \qquad y = 2$$

(b) $A = \begin{bmatrix} x^3 & 2 & 0 \\ 3 & y & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3x+1 & 2 & y \\ 3 & -2 & -1 \end{bmatrix}$

Solution:

(c) $A = \begin{bmatrix} 2 & -5 \\ 0 & x+y \\ 2y & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -5 \\ 0 & 2x+4 \\ y+1 & 1 \end{bmatrix}$

Solution:

4. Consider the following matrices:

$$A = \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -2 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & 1 \\ -2 & 2 & 4 \end{bmatrix}, \quad F = \begin{bmatrix} -2 & 1 & 3 \\ -1 & 1 & -2 \\ 0 & 1 & 3 \end{bmatrix}$$

Determine the following matrices, only if they are defined. If they are not defined, explain why.

A) $A + B$
B) $B + A$
C) $A - B$
D) $D + E$
E) $-A + 2B$
F) $A + C$
G) $A + D$
H) $A - A$
I) $C - C$
J) AB
K) AC
L) CA
M) AD
N) DA
O) EF
P) FE
Q) $A(3D)$
R) $(AC)D$
S) $A(CD)$
T) $(2A)(C(3D))$
U) $(2A)((2C)(2D))$
V) $A(CD + D)$
W) C^3

A) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$,

B) $\begin{bmatrix} 1 & 5 \\ -1 & 3 \\ -1 & -2 \end{bmatrix}$

C) $\begin{bmatrix} 3 & -1 \\ 7 & 3 \\ -1 & 0 \end{bmatrix}$

D) Not defined, because D is a 2×3 matrix and E is a 3×3 matrix, so they do not have the same dimensions.

E) $-1 \cdot \begin{bmatrix} 2 & 2 \\ 3 & 3 \\ -1 & -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} -1 & 3 \\ -4 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -4 & 5 \\ -11 & 3 \\ 0 & -2 \end{bmatrix}$

F) Not defined, because $A_{3\times 2} = [a_{ij}] = A$ and C is a $C_{2\times 2}$ matrix, so they do not have the same dimensions.

G) Not defined, because A is a 3×2 matrix and D is a 2×3 matrix, so they do not have the same dimensions.

H) $H = 0_{3x3}$

I) $H = 0_{2x2}$